

Challenge 2: AI Service Orchestrator for Informal Economy

Challenge Overview

Informal economy, including plumbers, electricians, tutors, beauticians, and home service providers, operates largely through:

- WhatsApp messages
- phone calls
- informal referrals

This results in:

- inefficient service matching
- missed opportunities
- lack of automation
- poor user experience

At the same time, users struggle to find:

- **reliable services quickly**
- **availability in real time**
- **trusted providers nearby**

Problem Statement

Build an **Agentic AI System** that automates the **end-to-end lifecycle of a service request** — from user intent to booking and follow-up.

Your system must:

1. Understand user service requests (in natural language)
2. Identify relevant providers using location/context
3. Select or recommend the best provider
4. Simulate booking and confirmation
5. Handle follow-up interactions
6. Show complete reasoning and workflow execution

Mandatory Requirement: Google Antigravity

Teams **MUST use Google Antigravity** as the core platform to:

- orchestrate agent workflows
- manage multi-step reasoning
- integrate tools (Maps, Search, APIs)
- execute actions (booking, notifications, etc.)

Use of external LLMs is allowed, but **Antigravity must be central to system logic and orchestration**.

Example User Scenario

User input (Roman Urdu / Urdu / English):

“Mujhe kal subah G-13 mein AC technician chahiye”

Expected Output

Service Request:

AC Technician

Location:

G-13

Time:

Tomorrow morning

Recommended Provider:

Ali AC Services (2.1 km away)

Reasoning:

Closest available provider with high rating

Simulated Booking:

- Slot booked: 10:00 AM

- Confirmation sent

Follow-up:

Reminder scheduled 1 hour before appointment

System Requirements

1. Intent Understanding

- Process natural language input
- Support:

§ Urdu

§ Roman Urdu

§ English

- Extract:

§ service type

§ location

§ time

2. Provider Discovery

- Use:

§ mock dataset OR

§ Google Maps / Places APIs

- Identify:

- § nearby providers
- § service category match

3. Matching & Ranking

- Rank providers based on:
 - § distance
 - § availability
 - § rating (simulated or real)
- Provide **clear reasoning for selection**

4. Decision & Recommendation

- Select best provider OR show top options
- Explain decision in simple terms

5. Action Simulation (CRITICAL REQUIREMENT)

System must simulate:

- booking confirmation
- provider assignment
- scheduling

Simulation can include:

- updating a mock booking system
- creating a confirmation message
- writing to a database/spreadsheet
- generating a booking receipt

6. Follow-Up Automation

- Simulate:
 - § reminders
 - § status updates
 - § completion confirmation

7. Agentic Workflow (MANDATORY)

System must demonstrate:

- multiple agents OR structured reasoning pipeline
- planning → decision → action → follow-up
- traceable logs of:
 - § decisions
 - § tool usage
 - § action execution

Deliverables

1. Working Prototype with Mobile App (MUST) and Web App (Optional)

2. Demo Video (3–5 minutes)

Must clearly show:

- user input
- system understanding
- provider matching
- booking simulation
- follow-up workflow

3. Agent Trace / Logs

- reasoning steps
- agent interactions
- action execution logs

4. Documentation (README)

Include:

- system architecture
- how Antigravity is used
- APIs/tools used
- assumptions and limitations

Evaluation Criteria

1. Use of Google Antigravity — 25%

- Core orchestration handled via Antigravity
- Effective use of tools (Maps, APIs)
- Demonstrates planning + execution

2. Agentic Reasoning & Workflow — 20%

- Multi-step reasoning
- Logical flow from request → decision → action
- Evidence of autonomy

3. Matching Quality & Decision Logic — 20%

- Relevant provider selection
- Clear ranking criteria
- Strong reasoning behind decisions

4. Action Simulation & Execution — 15%

- Booking process realistically simulated
- Clear system state change (confirmation, scheduling)
- End-to-end workflow demonstrated

5. Technical Implementation — 10%

- Clean architecture
- API/tool integration
- Robust handling of edge cases

6. Innovation & UX — 10%

- Creative approach
- Intuitive interface
- Clear and engaging demo

Important Guidelines

- This is NOT a simple listing or booking app
- Focus on **agentic automation, not UI complexity**
- At least one booking must be **simulated end-to-end**
- Must demonstrate **reasoning + decision-making**
- Use **mock data if real APIs are unavailable**
- Avoid use of real personal/sensitive data